Министерство образования Республики Беларусь
Учреждение образования
«Брестский государственный технический университет»
Кафедра ИИТ

Лабораторная работа №3
По дисциплине: «Обработка изображений в ИС»
Тема: «Обучение детекторов объектов»

Выполнил:
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Цель: осуществлять обучение нейросетевого детектора для решения задачи обнаружения заданных объектов

Ход работы

Общее задание

1. Базируясь на своем варианте, ознакомится с выборкой для обучения детектора, выполнить необходимые преобразования данных для организации процесса обучения (если это нужно!);
2. Для заданной архитектуры нейросетевого детектора организовать процесс обучения для своей выборки. Оценить эффективность обучения на тестовой выборке (mAP);
3. Реализовать визуализацию работы детектора из пункта 1 (обнаружение знаков на отдельных фотографиях из сети Интернет);
4. Оформить отчет по выполненной работе, залить исходный код и отчет в соответствующий репозиторий на github.

Задание по вариантам

В-т	**Детектор**	**Датасет**
11	YOLO12m	**Счетчики расхода воды:** https://universe.roboflow.com/koer3741-gmail-com/watermeteramrv2/dataset/1

Выборка	**Изображений**	**Объектов**
Обучающая (train)	3478	18304
Валидационная (valid)	389	2059
Тестовая (test)	389	2105

Код обучения и оценки

```
!pip install -q ultralytics roboflow
from pathlib import Path
import yaml
from roboflow import Roboflow
from ultralytics import YOLO

API_KEY = "<API_KEY>"
root = Path("/content/datasets/watermeteramrv2-v1")
if not root.exists():
    Roboflow(api_key=API_KEY).workspace("koer3741-gmail-
com").project("watermeteramrv2").version(1).download("yolov8", location=str(root))

cfg = yaml.safe_load((root / "data.yaml").read_text())
cfg.update(path=str(root), train="train/images", val="valid/images", test="test/images")
(root / "data.yaml").write_text(yaml.safe_dump(cfg))

model = YOLO("yolo12m.pt")
model.train(data=str(root / "data.yaml"), epochs=30, imgsz=640, batch=16, device=0,
            project="/content/runs", name="yolo12m_watermeter", patience=30, seed=42,
exist_ok=True)

best = YOLO(str(Path(model.trainer.save_dir) / "weights" / "best.pt"))
m = best.val(data=str(root / "data.yaml"), split="test", project="/content/runs", name="test",
exist_ok=True)
print("mAP50:", m.box.map50, "mAP50-95:", m.box.map, "P:", m.box.mp, "R:", m.box.mr)
```

Код визуализации

```
from google.colab import files
import cv2
import matplotlib.pyplot as plt

up = files.upload()
for name in up:
    r = best.predict(name, conf=0.25, verbose=False)[0]
    img = r.plot()
    cv2.imwrite("pred_" + name, img)
    plt.figure(figsize=(8, 6))
    plt.imshow(cv2.cvtColor(img, cv2.COLOR_BGR2RGB))
    plt.axis("off")
    plt.title(name)
    plt.show()
```

Результат:

```
Creating new Ultralytics Settings v0.0.8 file ☑
View Ultralytics Settings with 'yolo settings' or at
'/root/.config/Ultralytics/settings.json'
Update Settings with 'yolo settings key=value', i.e. 'yolo settings
runs_dir=path/to/dir'. For help see https://docs.ultralytics.com/usage/settings.
loading Roboflow workspace...
loading Roboflow project...
Downloading Dataset Version Zip in /content/datasets/watermeteramrv2-v1 to yolov8::
100%|██████████| 183798/183798 [00:03<00:00, 51677.59it/s]


Extracting Dataset Version Zip to /content/datasets/watermeteramrv2-v1 in yolov8::
100%|██████████| 8524/8524 [00:01<00:00, 7806.93it/s]
Downloading https://github.com/ultralytics/assets/releases/download/v8.4.0/yolo12m.pt
to 'yolo12m.pt': 100% ━━━━━━━━━━━━━━━━━━━━ 39.0MB 112.2MB/s 0.3s
Ultralytics 8.4.175 🚀 Python-3.13.15 torch-2.11.0+cu130 CUDA:0 (Tesla T4, 14913MiB)
engine/trainer: agnostic_nms=False, amp=True, angle=1.0, augment=False,
auto_augment=randaugment, batch=16, bgr=0.0, box=7.5, cache=False, cfg=None,
channels_last=None, classes=None, close_mosaic=10, cls=0.5, cls_pw=0.0,
cls_remap=True, compile=False, conf=None, copy_paste=0.0, copy_paste_mode=flip,
cos_lr=False, cutmix=0.0, data=/content/datasets/watermeteramrv2-v1/data.yaml,
degrees=0.0, deterministic=True, device=0, dfl=1.5, dgrad=0.5, dis=6.0,
distill_model=None, dlam=1.0, dlog=1.0, dnn=False, dropout=0.0, dynamic=False,
embed=None, epochs=30, erasing=0.4, exist_ok=True, fliplr=0.5, flipud=0.0,
format=torchscript, fraction=1.0, freeze=None, hsv_h=0.015, hsv_s=0.7, hsv_v=0.4,
imgsz=640, iou=0.7, kobj=1.0, line_width=None, lr0=0.01, lrf=0.01, mask_ratio=4,
max_det=300, mixup=0.0, mode=train, model=yolo12m.pt, momentum=0.937, mosaic=1.0,
multi_scale=0.0, name=yolo12m_watermeter, nbs=64, nms=None, opset=None,
optimize=False, optimizer=auto, overlap_mask=True, patience=30, perspective=0.0,
plots=True, pose=12.0, pretrained=True, profile=False, project=/content/runs,
quantize=None, rect=False, resume=False, retina_masks=False, rle=1.0, save=True,
save_conf=False, save_crop=False, save_dir=/content/runs/yolo12m_watermeter,
save_frames=False, save_json=False, save_period=-1, save_txt=False, scale=0.5,
seed=42, shear=0.0, show=False, show_boxes=True, show_conf=True, show_labels=True,
simplify=True, single_cls=False, source=None, split=val, stream_buffer=False,
task=detect, time=None, tracker=tracktrack.yaml, translate=0.1, val=True,
verbose=True, vid_stride=1, visualize=False, warmup_bias_lr=0.1, warmup_epochs=3.0,
warmup_momentum=0.8, weight_decay=0.0005, workers=8, workspace=None
Downloading https://ultralytics.com/assets/Arial.ttf to
'/root/.config/Ultralytics/Arial.ttf': 100% ━━━━━━━━━━━━━━━━━━━━ 755.1KB 22.7MB/s
0.0s
Overriding model.yaml nc=80 with nc=12

                   from  n    params  module
arguments
  0                  -1  1      1856  ultralytics.nn.modules.conv.Conv
[3, 64, 3, 2]
```

```
  1                  -1  1     73984  ultralytics.nn.modules.conv.Conv
[64, 128, 3, 2]
  2                  -1  1    111872  ultralytics.nn.modules.block.C3k2
[128, 256, 1, True, 0.25]
  3                  -1  1    590336  ultralytics.nn.modules.conv.Conv
[256, 256, 3, 2]
  4                  -1  1    444928  ultralytics.nn.modules.block.C3k2
[256, 512, 1, True, 0.25]
  5                  -1  1   2360320  ultralytics.nn.modules.conv.Conv
[512, 512, 3, 2]
  6                  -1  2   2689536  ultralytics.nn.modules.block.A2C2f
[512, 512, 2, True, 4]
  7                  -1  1   2360320  ultralytics.nn.modules.conv.Conv
[512, 512, 3, 2]
  8                  -1  2   2689536  ultralytics.nn.modules.block.A2C2f
[512, 512, 2, True, 1]
  9                  -1  1         0  torch.nn.modules.upsampling.Upsample
[None, 2, 'nearest']
 10             [-1, 6]  1         0  ultralytics.nn.modules.conv.Concat
[1]
 11                 -1  1   1248768  ultralytics.nn.modules.block.A2C2f
[1024, 512, 1, False, -1]
 12                 -1  1         0  torch.nn.modules.upsampling.Upsample
[None, 2, 'nearest']
 13             [-1, 4]  1         0  ultralytics.nn.modules.conv.Concat
[1]
 14                 -1  1    378624  ultralytics.nn.modules.block.A2C2f
[1024, 256, 1, False, -1]
 15                 -1  1    590336  ultralytics.nn.modules.conv.Conv
[256, 256, 3, 2]
 16            [-1, 11]  1         0  ultralytics.nn.modules.conv.Concat
[1]
 17                 -1  1   1183232  ultralytics.nn.modules.block.A2C2f
[768, 512, 1, False, -1]
 18                 -1  1   2360320  ultralytics.nn.modules.conv.Conv
[512, 512, 3, 2]
 19             [-1, 8]  1         0  ultralytics.nn.modules.conv.Concat
[1]
 20                 -1  1   1642496  ultralytics.nn.modules.block.C3k2
[1024, 512, 1, True]
 21        [14, 17, 20]  1   1420276  ultralytics.nn.modules.head.Detect
[12, 16, None, [256, 512, 512]]
YOLOv12m summary: 292 layers, 20,146,740 parameters, 20,146,724 gradients, 71.1
GFLOPs

Transferred 745/751 items from pretrained weights
Freezing layer 'model.21.dfl.conv.weight'
AMP: running Automatic Mixed Precision (AMP) checks...
AMP: downloading yolo26n.pt for AMP checks (one-time, not used for training)...
Downloading https://github.com/ultralytics/assets/releases/download/v8.4.0/yolo26n.pt
to 'weights/yolo26n.pt': 100% ━━━━━━━━━━━━━━━━━━━━ 5.3MB 94.5MB/s 0.1s
AMP: checks passed ☑
train: Fast image access ☑ (ping: 0.0±0.0 ms, read: 1339.2±679.0 MB/s, size: 41.7
KB)
train: Scanning /content/datasets/watermeteramrv2-v1/train/labels... 3478 images, 0
backgrounds, 0 corrupt: 100% ━━━━━━━━━━━━━━━━━━━━ 3478/3478 1.9Kit/s 1.8s
train: New cache created: /content/datasets/watermeteramrv2-v1/train/labels.cache
albumentations: Blur(p=0.01, blur_limit=(3, 7)), MedianBlur(p=0.01, blur_limit=(3,
7)), ToGray(p=0.01, method='weighted_average', num_output_channels=3), CLAHE(p=0.01,
clip_limit=(1.0, 4.0), tile_grid_size=(8, 8))
val: Fast image access ☑ (ping: 0.0±0.0 ms, read: 765.7±352.7 MB/s, size: 43.4 KB)
val: Scanning /content/datasets/watermeteramrv2-v1/valid/labels... 389 images, 0
backgrounds, 0 corrupt: 100% ━━━━━━━━━━━━━━━━━━━━ 389/389 1.5Kit/s 0.3s
```

```
val: New cache created: /content/datasets/watermeteramrv2-v1/valid/labels.cache
optimizer: 'optimizer=auto' found, ignoring 'lr0=0.01' and determining best
'optimizer' and 'lr0' automatically...
optimizer: AdamW(lr=0.000625, momentum=0.9) with parameter groups 123
weight(decay=0.0), 130 weight(decay=0.0005), 129 bias(decay=0.0)
Plotting labels to /content/runs/yolo12m_watermeter/labels.jpg...
Using 3478 train, 389 val images for fraction=1.0 at imgsz=640
Using 2 dataloader workers
Logging results to /content/runs/yolo12m_watermeter
Starting training for 30 epochs...

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
       1/30      11.4G      1.473      1.727      1.213         40        640: 100%
──────────────────── 218/218 1.5it/s 2:24
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-
95): 100% ──────────────────── 13/13 1.6it/s 8.3s
                   all        389       2059      0.662      0.679       0.71
0.329

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
       2/30      11.2G      1.431      1.041      1.164         51        640: 100%
──────────────────── 218/218 1.6it/s 2:19
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-
95): 100% ──────────────────── 13/13 1.8it/s 7.1s
                   all        389       2059      0.782      0.793      0.859
0.447

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
       3/30      11.2G      1.446       0.96      1.171         42        640: 100%
──────────────────── 218/218 1.6it/s 2:19
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-
95): 100% ──────────────────── 13/13 1.7it/s 7.5s
                   all        389       2059      0.822      0.791      0.853
0.457

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
       4/30      11.2G      1.395     0.8715      1.149         81        640: 100%
──────────────────── 218/218 1.6it/s 2:19
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-
95): 100% ──────────────────── 13/13 1.7it/s 7.5s
                   all        389       2059      0.808      0.805      0.866
0.478

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
       5/30      11.2G      1.384     0.8426      1.143         51        640: 100%
──────────────────── 218/218 1.6it/s 2:19
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-
95): 100% ──────────────────── 13/13 1.8it/s 7.1s
                   all        389       2059      0.888      0.844       0.91
0.503

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
       6/30      11.2G      1.362     0.7921      1.135         70        640: 100%
──────────────────── 218/218 1.6it/s 2:19
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-
95): 100% ──────────────────── 13/13 1.8it/s 7.2s
                   all        389       2059      0.905       0.89      0.925
0.49

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
       7/30      11.2G       1.35     0.7516      1.123         57        640: 100%
──────────────────── 218/218 1.6it/s 2:18
```

```
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-
95): 100% ━━━━━━━━━━━━━━━━━━━━ 13/13 1.8it/s 7.2s
                   all        389       2059      0.903      0.883      0.921
0.464

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
       8/30      11.2G      1.318     0.7212      1.114         51        640: 100%
━━━━━━━━━━━━━━━━━━━━ 218/218 1.6it/s 2:18
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-
95): 100% ━━━━━━━━━━━━━━━━━━━━ 13/13 1.8it/s 7.2s
                   all        389       2059      0.912      0.907      0.943
0.528

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
       9/30      11.2G      1.313     0.7068      1.108         59        640: 100%
━━━━━━━━━━━━━━━━━━━━ 218/218 1.6it/s 2:18
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-
95): 100% ━━━━━━━━━━━━━━━━━━━━ 13/13 1.8it/s 7.3s
                   all        389       2059      0.921      0.897      0.947
0.537

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      10/30      11.2G      1.308     0.6893      1.104         42        640: 100%
━━━━━━━━━━━━━━━━━━━━ 218/218 1.6it/s 2:19
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-
95): 100% ━━━━━━━━━━━━━━━━━━━━ 13/13 1.7it/s 7.6s
                   all        389       2059      0.928      0.921       0.95
0.515

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      11/30      11.2G      1.303     0.6826      1.099         32        640: 100%
━━━━━━━━━━━━━━━━━━━━ 218/218 1.6it/s 2:18
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-
95): 100% ━━━━━━━━━━━━━━━━━━━━ 13/13 1.7it/s 7.5s
                   all        389       2059      0.908      0.896      0.941
0.507

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      12/30      11.2G      1.287     0.6604        1.1         49        640: 100%
━━━━━━━━━━━━━━━━━━━━ 218/218 1.6it/s 2:18
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-
95): 100% ━━━━━━━━━━━━━━━━━━━━ 13/13 1.7it/s 7.4s
                   all        389       2059      0.931      0.941      0.956
0.515

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      13/30      11.2G      1.267      0.644      1.086         27        640: 100%
━━━━━━━━━━━━━━━━━━━━ 218/218 1.6it/s 2:18
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-
95): 100% ━━━━━━━━━━━━━━━━━━━━ 13/13 1.8it/s 7.4s
                   all        389       2059      0.946      0.931      0.963
0.538

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      14/30      11.2G      1.267     0.6369      1.086         29        640: 100%
━━━━━━━━━━━━━━━━━━━━ 218/218 1.6it/s 2:18
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-
95): 100% ━━━━━━━━━━━━━━━━━━━━ 13/13 1.8it/s 7.1s
                   all        389       2059       0.93      0.927      0.958
0.556
```

```
      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      15/30      11.2G      1.247     0.6185      1.086         78        640: 100%
━━━━━━━━━━━━━━━━━━━━ 218/218 1.6it/s 2:18
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-
95): 100% ━━━━━━━━━━━━━━━━━━━━ 13/13 1.8it/s 7.1s
                   all        389       2059      0.936      0.923      0.957
0.542

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      16/30      11.2G       1.26     0.6175      1.082         74        640: 100%
━━━━━━━━━━━━━━━━━━━━ 218/218 1.6it/s 2:18
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-
95): 100% ━━━━━━━━━━━━━━━━━━━━ 13/13 1.8it/s 7.3s
                   all        389       2059      0.941      0.929      0.965
0.552

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      17/30      11.2G      1.235     0.5996      1.074         45        640: 100%
━━━━━━━━━━━━━━━━━━━━ 218/218 1.6it/s 2:18
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-
95): 100% ━━━━━━━━━━━━━━━━━━━━ 13/13 1.8it/s 7.4s
                   all        389       2059      0.936      0.934      0.966
0.536

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      18/30      11.2G      1.225     0.5873      1.072         60        640: 100%
━━━━━━━━━━━━━━━━━━━━ 218/218 1.6it/s 2:18
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-
95): 100% ━━━━━━━━━━━━━━━━━━━━ 13/13 1.8it/s 7.2s
                   all        389       2059      0.946      0.941      0.968
0.528

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      19/30      11.2G      1.228     0.5791      1.072         61        640: 100%
━━━━━━━━━━━━━━━━━━━━ 218/218 1.6it/s 2:18
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-
95): 100% ━━━━━━━━━━━━━━━━━━━━ 13/13 1.8it/s 7.1s
                   all        389       2059      0.955      0.948      0.973
0.584

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      20/30      11.2G      1.208     0.5793      1.066         31        640: 100%
━━━━━━━━━━━━━━━━━━━━ 218/218 1.6it/s 2:18
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-
95): 100% ━━━━━━━━━━━━━━━━━━━━ 13/13 1.8it/s 7.2s
                   all        389       2059      0.953      0.929      0.971
0.567
Closing dataloader mosaic
albumentations: Blur(p=0.01, blur_limit=(3, 7)), MedianBlur(p=0.01, blur_limit=(3,
7)), ToGray(p=0.01, method='weighted_average', num_output_channels=3), CLAHE(p=0.01,
clip_limit=(1.0, 4.0), tile_grid_size=(8, 8))

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      21/30      11.2G      1.202     0.5127      1.113         30        640: 100%
━━━━━━━━━━━━━━━━━━━━ 218/218 1.6it/s 2:19
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-
95): 100% ━━━━━━━━━━━━━━━━━━━━ 13/13 1.8it/s 7.1s
                   all        389       2059      0.944      0.956      0.974
0.589

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
```

```
      22/30      11.2G      1.185     0.4978      1.106         31        640: 100%
━━━━━━━━━━━━ 218/218 1.6it/s 2:17
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-
95): 100% ━━━━━━━━━━━━ 13/13 1.7it/s 7.5s
                   all        389       2059      0.949      0.944      0.973
0.576

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      23/30      11.2G      1.165     0.4912      1.093         28        640: 100%
━━━━━━━━━━━━ 218/218 1.6it/s 2:17
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-
95): 100% ━━━━━━━━━━━━ 13/13 1.8it/s 7.2s
                   all        389       2059      0.953      0.949      0.975
0.582

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      24/30      11.2G      1.148     0.4772      1.083         32        640: 100%
━━━━━━━━━━━━ 218/218 1.6it/s 2:18
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-
95): 100% ━━━━━━━━━━━━ 13/13 1.8it/s 7.1s
                   all        389       2059      0.958      0.951      0.978
0.589

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      25/30      11.2G       1.13     0.4718      1.072         32        640: 100%
━━━━━━━━━━━━ 218/218 1.6it/s 2:17
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-
95): 100% ━━━━━━━━━━━━ 13/13 1.8it/s 7.4s
                   all        389       2059       0.96      0.946      0.978
0.582

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      26/30      11.2G      1.116     0.4573      1.063         30        640: 100%
━━━━━━━━━━━━ 218/218 1.6it/s 2:18
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-
95): 100% ━━━━━━━━━━━━ 13/13 1.8it/s 7.2s
                   all        389       2059      0.964      0.949      0.979
0.604

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      27/30      11.2G       1.11     0.4494      1.063         32        640: 100%
━━━━━━━━━━━━ 218/218 1.6it/s 2:18
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-
95): 100% ━━━━━━━━━━━━ 13/13 1.8it/s 7.1s
                   all        389       2059      0.962      0.952       0.98
0.603

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      28/30      11.2G      1.094     0.4456      1.056         33        640: 100%
━━━━━━━━━━━━ 218/218 1.6it/s 2:18
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-
95): 100% ━━━━━━━━━━━━ 13/13 1.7it/s 7.5s
                   all        389       2059      0.964      0.947       0.98
0.603

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      29/30      11.2G      1.083      0.437      1.054         30        640: 100%
━━━━━━━━━━━━ 218/218 1.6it/s 2:18
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-
95): 100% ━━━━━━━━━━━━ 13/13 1.8it/s 7.1s
```

```
                   all        389       2059      0.963      0.956      0.981
0.605

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      30/30      11.2G      1.064     0.4301      1.046         33        640: 100%
━━━━━━━━━━━━ 218/218 1.6it/s 2:18
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-
95): 100% ━━━━━━━━━━━━ 13/13 1.8it/s 7.0s
                   all        389       2059      0.964      0.955      0.982
0.609

30 epochs completed in 1.247 hours.
Optimizer stripped from /content/runs/yolo12m_watermeter/weights/last.pt, 40.8MB
Optimizer stripped from /content/runs/yolo12m_watermeter/weights/best.pt, 40.8MB

Validating /content/runs/yolo12m_watermeter/weights/best.pt...
Ultralytics 8.4.175 🚀 Python-3.13.15 torch-2.11.0+cu130 CUDA:0 (Tesla T4, 14913MiB)
YOLOv12m summary (fused): 169 layers, 20,114,164 parameters, 0 gradients, 70.4 GFLOPs
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-
95): 100% ━━━━━━━━━━━━ 13/13 1.4it/s 9.5s
                   all        389       2059      0.964      0.955      0.982
0.61
                     0        151        196      0.898      0.851       0.93
0.545
                     1        122        161      0.923       0.95      0.974
0.588
                     2         97        106      0.971      0.943      0.979
0.55
                     3        124        148      0.969      0.953      0.989
0.639
                     4        148        184      0.995      0.978       0.99
0.633
                     5         96        111      0.951      0.955      0.987
0.567
                     6        125        151      0.988       0.96      0.991
0.598
                     7         98        124       0.99      0.952      0.973
0.598
                     8         60         67       0.97      0.965      0.989
0.608
                     9         58         60      0.967      0.971      0.992
0.598
               counter        367        373      0.982      0.995      0.995
0.803
                liters        375        378      0.966      0.989      0.991
0.588
Speed: 1.2ms preprocess, 16.2ms inference, 0.0ms loss, 1.9ms postprocess per image
Results saved to /content/runs/yolo12m_watermeter
Ultralytics 8.4.175 🚀 Python-3.13.15 torch-2.11.0+cu130 CUDA:0 (Tesla T4, 14913MiB)
YOLOv12m summary (fused): 169 layers, 20,114,164 parameters, 0 gradients, 70.4 GFLOPs
WARNING ⚠ val: Slow image access detected (ping: 0.0±0.0 ms, read: 20.1±10.8 MB/s,
size: 54.4 KB). Use local storage instead of remote/mounted storage for better
performance. See https://docs.ultralytics.com/guides/model-training-tips
val: Scanning /content/datasets/watermeteramrv2-v1/test/labels... 389 images, 0
backgrounds, 0 corrupt: 100% ━━━━━━━━━━━━ 389/389 860.5it/s 0.5s
val: New cache created: /content/datasets/watermeteramrv2-v1/test/labels.cache
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-
95): 100% ━━━━━━━━━━━━ 25/25 1.5it/s 17.1s
                   all        389       2105      0.961      0.946      0.972
0.589
                     0        162        213      0.918      0.898      0.947
0.528
```

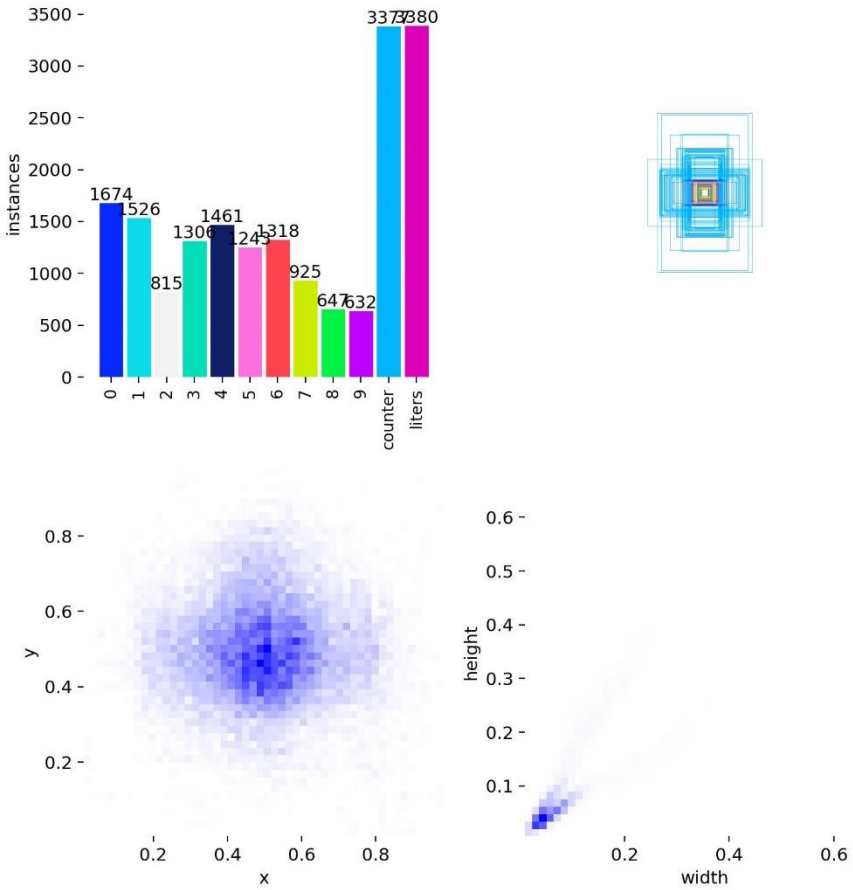
```
                   1        108        153       0.907       0.953       0.944
0.531
                   2         87         90       0.956       0.957       0.962
0.538
                   3        125        154       0.978       0.883       0.956
0.581
                   4        132        165       0.973       0.982       0.993
0.611
                   5        119        167       0.968       0.958       0.987
0.556
                   6        112        130       0.976       0.949       0.981
0.6
                   7        100        115       0.988       0.965       0.994
0.595
                   8         64         75       0.951       0.907        0.96
0.562
                   9         70         75       0.958       0.918       0.958
0.599
             counter        371        383       0.987       0.999       0.991
0.792
              liters        377        385       0.967       0.987       0.986
0.577
Speed: 1.9ms preprocess, 36.4ms inference, 0.0ms loss, 1.2ms postprocess per image
Results saved to /content/runs/test
mAP50: 0.9715837599770225 mAP50-95: 0.5891618948706598 P: 0.9605234661027476 R:
0.9462239909459039
```

Выбрать файлы schetchik-960x540.jpg

schetchik-960x540.jpg(image/jpeg) - 80760 bytes, last modified: 13.10.2026 - 100% done

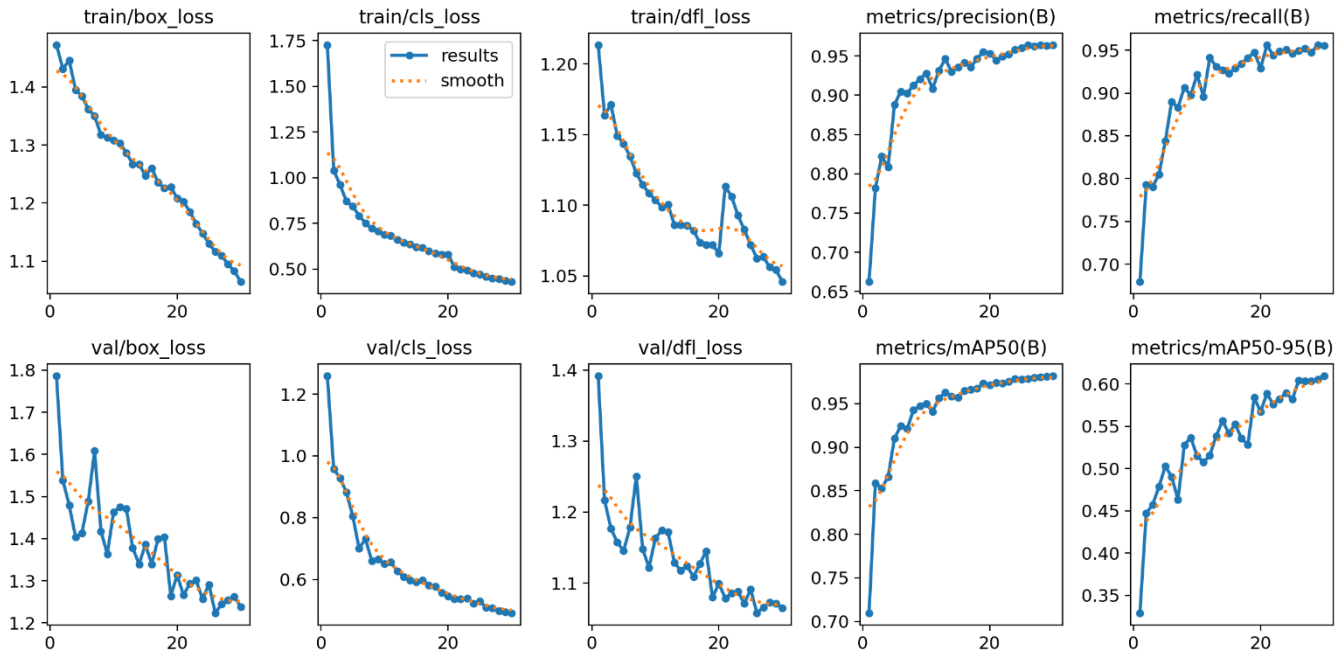
Saving schetchik-960x540.jpg to schetchik-960x540.jpg





Распределение классов, расположения и размеров рамок в обучающей выборке

Эпоха	**mAP@50**	**mAP@50-95**
1	0,710	0,329
2	0,859	0,447
3	0,853	0,457
4	0,866	0,478
5	0,910	0,503
10	0,950	0,515
15	0,957	0,542
20	0,971	0,567
25	0,978	0,582
30	0,982	0,609



Графики потерь и метрик в процессе обучения

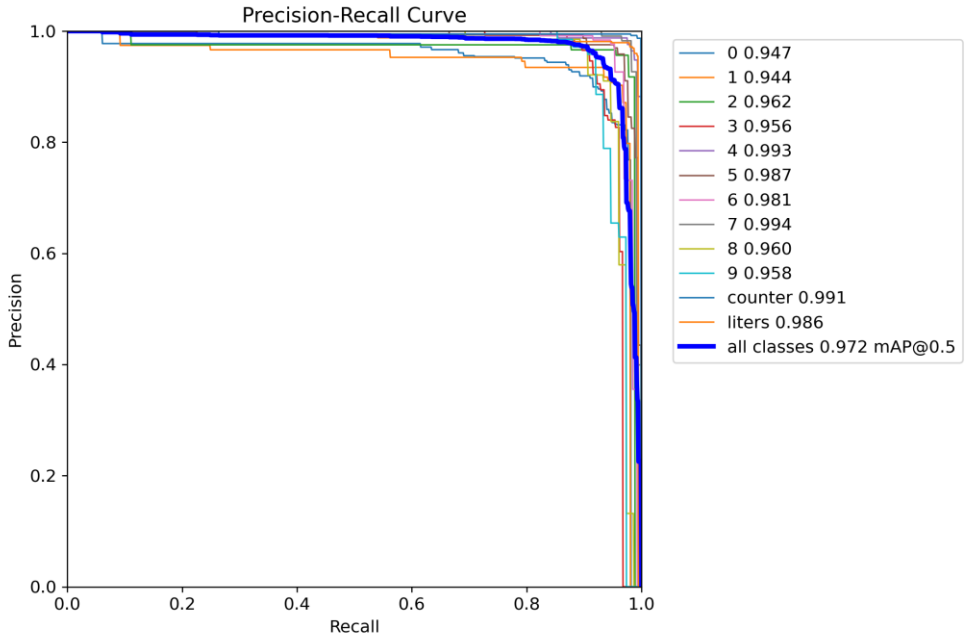
Итоговая оценка выполнена на тестовой выборке (389 изображений, 2105 объектов). Метрики на тестовой выборке: mAP@50 = 0,9716, mAP@50-95 = 0,5892, Precision = 0,9605, Recall = 0,9462. Близкие значения на валидационной (mAP@50 = 0,982) и тестовой выборках говорят об отсутствии заметного переобучения.

Выборка	**Precision**	**Recall**	**mAP@50**	**mAP@50-95**
Валидационная	0,964	0,955	0,982	0,610
Тестовая	0,961	0,946	0,972	0,589

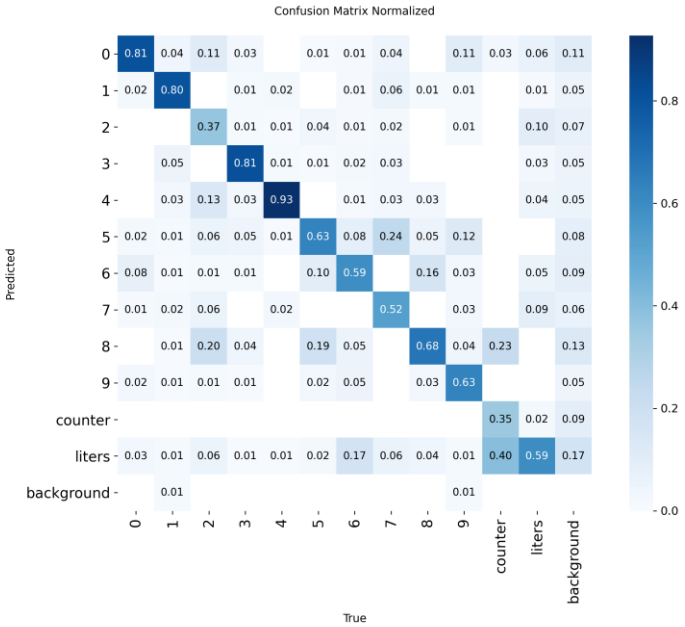
Результаты по классам на тестовой выборке приведены в таблице. Лучше всего модель определяет класс «counter» (mAP@50-95 = 0,792) — это крупный объект простой формы. Точность локализации цифр ниже (mAP@50-95 от 0,53 до 0,61), поскольку они имеют малый размер: небольшое смещение рамки заметно снижает IoU при высоких порогах. Наименьший mAP@50 у цифр «1» (0,944) и «0» (0,947).

Класс	**Precision**	**Recall**	**mAP@50**	**mAP@50-95**
0	0.918	0.898	0.947	0.528
1	0.907	0.953	0.944	0.531
2	0.956	0.957	0.962	0.538
3	0.978	0.883	0.956	0.581
4	0.973	0.982	0.993	0.611
5	0.968	0.958	0.987	0.556
6	0.976	0.949	0.981	0.600
7	0.988	0.965	0.994	0.595
8	0.951	0.907	0.960	0.562

Класс	Precision	Recall	mAP@50	mAP@50-95
9	0.958	0.918	0.958	0.599
counter	0.987	0.999	0.991	0.792
liters	0.967	0.987	0.986	0.577
все классы	0.961	0.946	0.972	0.589



Кривые Precision-Recall по классам (тестовая выборка)



Нормированная матрица ошибок



Примеры предсказаний детектора на тестовой выборке

Вывод: осуществлять обучение нейросетевого детектора для решения задачи обнаружения заданных объектов